

Week 4 Problem Set

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1. Start a new document, `problemset4.Rmd`, in your `soc10problemsets` repository. (You can download a template here, but be sure to save it in your own `soc10problemsets` repo). In the chunk below, show how you would gather data on California's ethnoracial demographics for 2020 using `tidycensus`. (Note: in the chunk below, `eval` is set to `false`, so you won't see any results. That is ok! Just show the code. And remove your personal key). **Commit** your changes.

To gather California's ethnoracial demographics for 2020, I used the `tidycensus` package which provides a way to access U.S. Census data directly in R and I chose `get_decennial()` over `get_acs()` because the decennial census is conducted every 10 years and provides exact population counts rather than estimates with margins of error, making it the most accurate source for racial demographic data for 2020.

I selected five racial variables from the 2020 Decennial Census demographic file: White alone (`P2_005N`), Black or African American alone (`P2_006N`), Asian alone (`P2_008N`), Hispanic or Latino (`P2_002N`), and Native American (`P2_007N`). I chose these five groups because they represent the major ethnoracial groups in California and align with the racial categories used in the `ca_dem_table` data we worked with in Problem Set 3, allowing for consistency across assignments. I used `geography = "county"` to get a breakdown by California county, which provides more exact demographic information than a single statewide estimate.

```
library(tidycensus)
library(tidyverse)

census_api_key("my_key")

# ca ethnoracial demographics for 2020
ca_race <- get_decennial(
  geography = "county",
  variables = c(
    White = "P2_005N",
    Black = "P2_006N",
    Asian = "P2_008N",
    Hispanic = "P2_002N",
    Native_American = "P2_007N"
  ),
  state = "CA",
  year = 2020,
  sumfile = "dhc"
)
```

2. In *Bit by Bit*, Salganik describes several possibilities and challenges of survey data in the digital age. Do you think any of these have changed since 2017 (when the book was written)? Pick a new type of survey or data collection, and describe its current possibilities and limitations. **Commit** your changes.

In Bit by Bit, Salganik discusses the rise of smartphone surveys as a new method for reaching hard-to-reach populations and collecting real-time data and one example he highlights is the study of formerly incarcerated men, where smartphones were used to administer daily surveys tracking well-being and employment searches after prison release which we discussed in class. While this was groundbreaking in 2017, digital data collection has shifted dramatically due to tech and ai advancements.

One major change is the rise of data collection through wearable technology like Apple Watches, Fitbits, and Oura Rings. Rather than asking people questions, researchers can now continuously collect biometric data like heart rate, sleep patterns, stress levels, and physical activity without any active participation from the respondent after initial consent. Stanford researchers are even currently conducting a study on Oura Ring usage on college students. One possibility this has is that since this method is always-on and non-reactive in the same way Salganik describes big data, people often forget they are being monitored and behave naturally so therefore wearable tech can capture data that people cannot accurately self-report, like exact sleep duration or stress responses to daily events, which traditional surveys rely on memory and self-perception for. This opens up entirely new research questions in medical sociology, mental health research, and the study of inequality in physical health outcomes. A limitation is wearables are expensive and disproportionately owned by wealthier and younger populations, making samples highly non-representative and therefore, like the Stanford study, require the researchers to provide the wearable technology as its not frequent people own it already. The data is also extremely sensitive since information collected from these is deeply personal and raises serious ethical concerns around privacy, consent, and data security.

3. Gather the data from this Wikipedia page on incarceration in the US (you can use the code from Section 4 of the course website to gather this). Use `mutate()`, `parse_number()` from the `readr` package, and the magrittr pipe (`%<>%`) to fix the issue with the `Count` variable. Then plot the `Count` variable over time with `ggplot()`. **Commit** your changes.

I gathered the incarceration data from Wikipedia using `rvest` and then used `mutate()` and `parse_number()` to fix the `Count` variable, which was stored as a character with commas rather than as a numeric value. The plot shows that incarceration counts in the US increased dramatically from 1940 through the 2000s before beginning to decline slightly in recent years.

```
library(rvest)
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
library(magrittr)
library(readr)
library(ggplot2)

us_inc <- read_html("https://en.wikipedia.org/wiki/Incarceration_in_the_United_States")

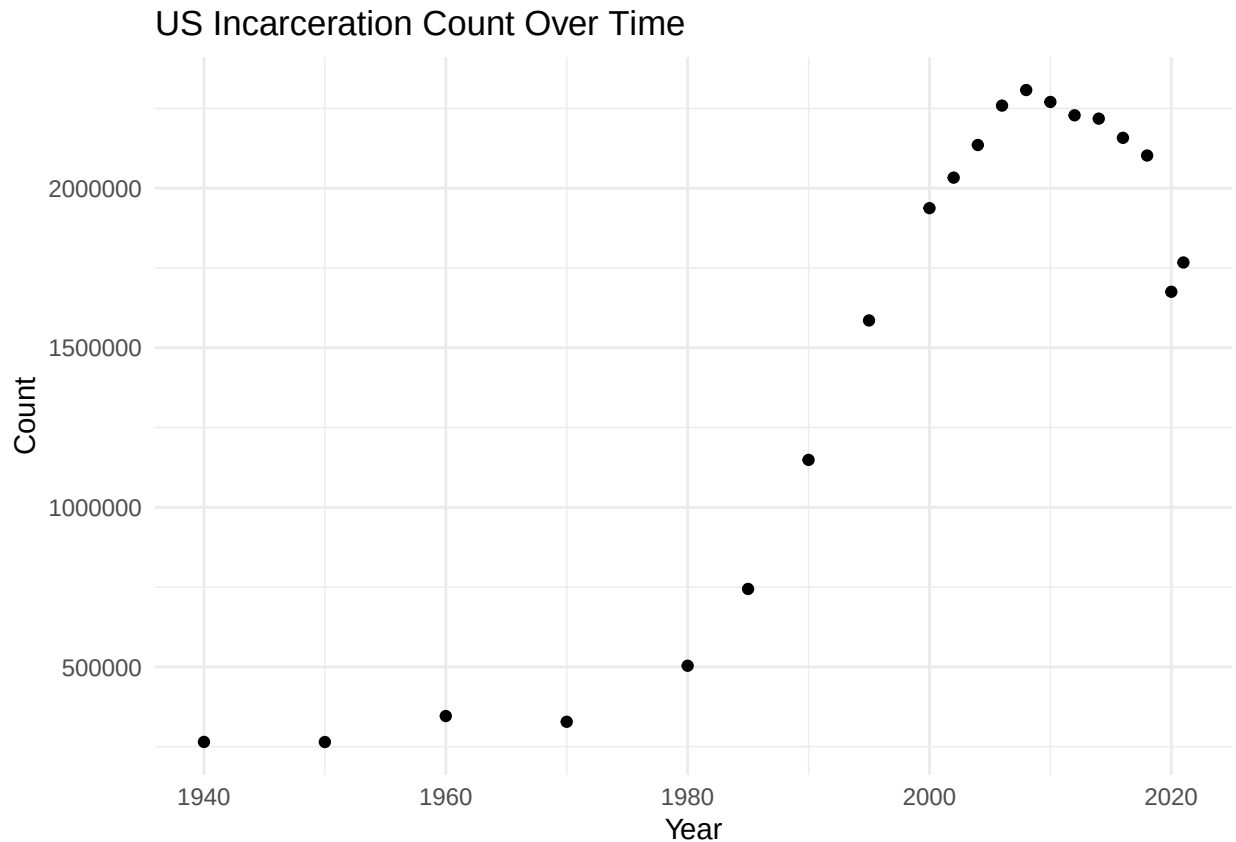
us_inc_table <- html_nodes(us_inc,
                           xpath = '//*[@id="mw-content-text"]/div[2]/div[5]/table')

us_inc_table %<>%
  html_table()

us_inc_table %<>%
  as.data.frame()

us_inc_table %<>%
  mutate(Count = parse_number(Count))
```

```
ggplot(us_inc_table, aes(x = Year, y = Count)) +
  geom_point() +
  labs(title = "US Incarceration Count Over Time",
       x = "Year",
       y = "Count") +
  theme_minimal()
```

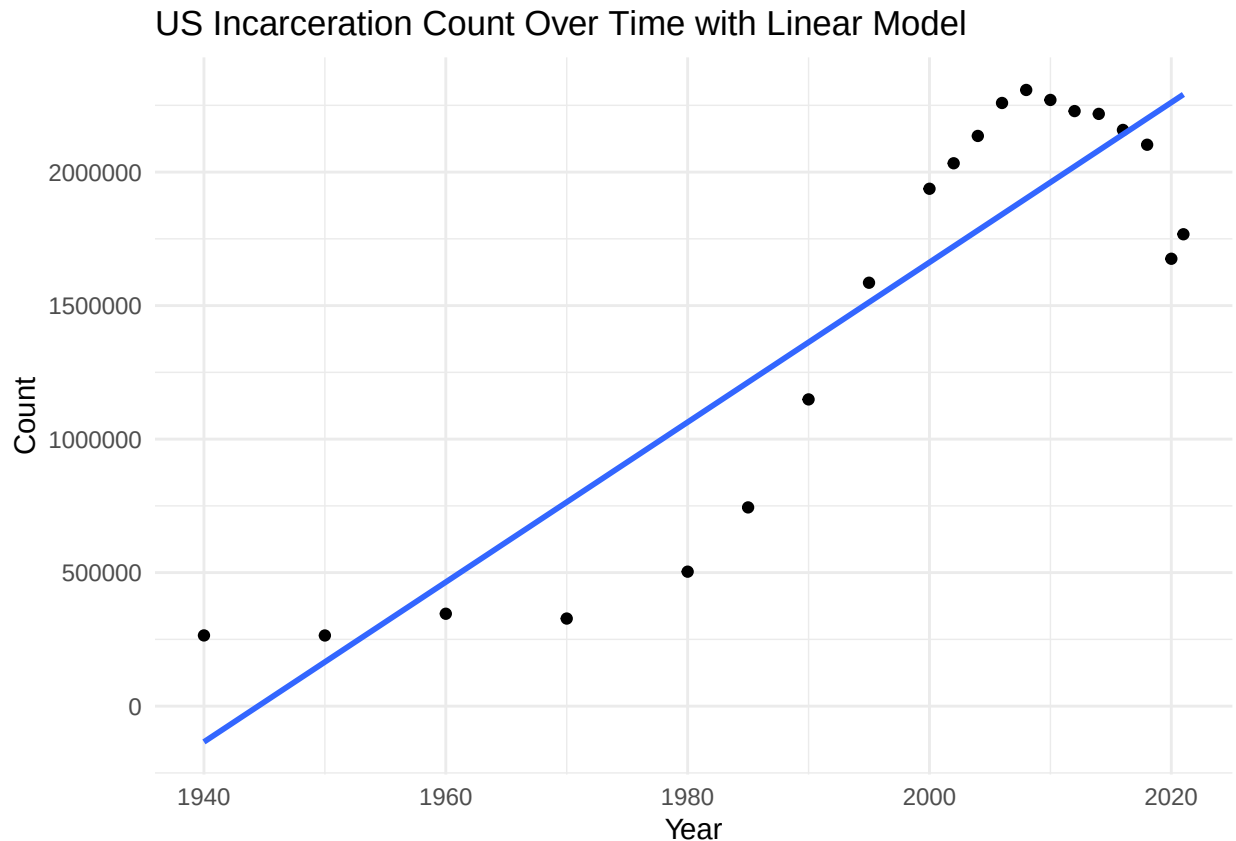


- Now model it! Add a linear model of incarceration over time to your plot. (Note: If you are unable to plot question 3 correctly, you can plot **Rate** as the outcome for this question). **Commit** your changes.

I added a linear model to my incarceration plot from Q3 using `geom_smooth()` with `method = "lm"` which fits a line of best fit through the data points. I can see from the plot that the linear model shows an overall upward trend in incarceration counts over time, however I notice it does not perfectly capture what is actually happening in the data. The counts rose very sharply through the 1980s and 1990s and then began to decline after around 2008, which a straight line cannot fully represent. This makes me think a non-linear model might actually fit this data better, which I will explore in Q5.

```
ggplot(us_inc_table, aes(x = Year, y = Count)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "US Incarceration Count Over Time with Linear Model",
       x = "Year",
       y = "Count") +
  theme_minimal()
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



5. Would a linear or non-linear model be more appropriate for explaining and predicting trends in US society and incarceration? Explain the benefits and drawbacks of each. **Commit** your changes.

Looking at the incarceration data, I think a non-linear model is more appropriate because the count rose sharply from around 264,000 in 1940 to over 2.3 million in 2008 and then declined to around 1.7 million by 2020. That kind of curve with a clear peak cannot be captured by a straight line without removing key data points and necessary info.

A linear model has the benefit of being simple and easy to interpret because the slope gives a clear number for how much incarceration changes per year, which makes it easy to communicate findings and make basic predictions. However the drawback is that it assumes a constant rate of change, which does not match this data at all. A linear model would predict incarceration keeps rising indefinitely, which we can see from the data is not what actually happened after 2008.

A non-linear model like the model we used in class has the benefit of following the actual shape of the data much more closely, capturing both the rapid rise through the 1990s and the decline after 2008. This gives a much more honest and accurate description of the trend. The drawback is that non-linear models are harder to interpret since there is no single slope value, and they can overfit the data by following small bumps and dips too closely, which makes future predictions less reliable and harder to determine.